

PAPER_1368 — Controlled Nuclear Fusion (Lawson Criterion)

Framework: UQFF — Star-Magic v5.27+ | **Tier:** H Applied/Engineering (CLOSED) **Calculator:** `calculate_paradox({"paradox": "fusion_lawson"})`

Master Expression

$\text{Lawson_UQFF} = 3e21 / K_MEX = 1.44e21 \text{ keV}\cdot\text{s}/\text{m}^3$

Match: reduces by factor K_MEX

Interpretation

SCm phonon-assisted confinement reduces ignition threshold by $K_MEX = 2.083\times$. ITER target achievable.

UQFF Locked Primitives

- $D_phys = 4$, $D_BSFG = 6$, $D_crit = 26$, $SO_5 = 10$, $A_5 = 60$, $N_CH = 9$
- $K_MEX = 25/12$, $\beta_i = 0.6029$, $\Phi_res = 0.84$, $\Lambda = 0.00729735$, $F_TRZ = 0.1$
- $\omega_SCm = 1.25 \text{ THz}$ ($\alpha = \Lambda$ closed ledger identity)

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, Youngstown, OH.